Inquiry Evolution & Discovery Engine

A step-by-step record of how the conversation advanced from prompts, to theories, to falsification, to a research methodology.

> The objective is not to produce beautiful answers. The objective is to build a process that becomes more reliable each time reality proves it wrong.

- Format: step-by-step progression, prompt templates, lessons, tests, and next actions.
- Use: apply this to any domain - AI, business, science, history, philosophy, software, economics, or personal research.
- Core standard: every idea must move from elegance toward evidence, prediction, and revision.

1. The Arc: What Happened in the Conversation

The conversation did not merely generate prompts. It evolved through levels of abstraction, each one correcting the previous one.

Stage	Question	What It Produced	Risk Discovered
1. Project planning	Can a very smart model find the hidden structure of a project?	Project decomposition with hidden architecture	It may only make a better plan, not a better proble
2. Opportunity discovery	Can it find a common, overlooked problem?	Agent accountability / trust infrastructure	Narrative risk: plausible but maybe not real deman
3. Falsification	Can it destroy its own thesis?	Stronger, revised thesis: agent acceptance network	The first thesis had weak analogies and timing issu
4. Civilization theory	What assumptions about civilization are failing?	Scarce attention vs costly verification	Grand theory risk: elegant but hard to test.
5. Reality audit	What does current evidence say?	Evidence ledger, confidence scores, and claim downgraded or contradicted.	Stated predictions
6. Discovery engine	How are discoveries made?	Compress -> predict -> reality veto	No universal discovery algorithm; only a universal c
7. Research program	How can humans and AI discover better together?	Agent-augmented epistemology	The method itself must be tested.

> We began by trying to get a better answer from AI. We ended by designing a better way to test answers from AI.

2. Step-by-Step Prompts That Advanced the Thinking

These are the core prompts, ordered by how the conversation advanced. Each one has a different job. Do not use them all at once. Use them as stages.

Prompt 1 - Hidden Project Architecture

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You are not acting as a coder or project manager. You are acting as a systems thinker.

Assume the project I describe has a hidden architecture that makes it dramatically simpler once
    discovered.
Your task is not to execute the project. Your task is to discover the decomposition that makes
    execution almost inevitable.

For the project:
1. Restate the real problem, not only the stated request.
2. Identify the core invariant.
3. Identify irreducible abstractions.
4. Separate essential constraints from accidental complexity.
5. Find hidden assumptions.
6. Construct a dependency graph instead of a linear roadmap.
7. Identify the 5% of decisions that determine 95% of future complexity.
8. Attack your own architecture.
9. Revise until the project cannot be made conceptually simpler without losing capability.

Output a foundation document that future AI or human implementers can continue from.
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Prompt 2 - Single-Shot Opportunity Discovery

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You are one of the most capable interdisciplinary thinkers imaginable.

Your objective is to identify one opportunity with the highest combination of long-term impact,
    feasibility, defensibility, scalability, economic value, societal value, and originality of
    insight.

Do not generate many startup ideas. Internally evaluate many possibilities, then select only the
    strongest candidate.

For the selected opportunity:
1. Explain why it wins.
2. Explain why alternatives were rejected.
3. Identify the hidden assumption most people are missing.
4. Reconstruct the problem from first principles.
5. Discover the smallest conceptual architecture.
6. Produce a complete product blueprint.
7. Produce a dependency graph.
8. Stress test the thesis.
9. Produce an AI handoff document for future continuation.

Prioritize truth over novelty. If the idea is not strong, say so.
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Prompt 3 - Scientific Critique and Falsification

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You previously selected a thesis. Do not defend it.

Your task is to behave like a panel of world-class critics trying to destroy the thesis if it
    deserves to be destroyed.

1. Steelman the opposition.
2. Identify hidden assumptions.
3. Find structural weaknesses.
4. Design kill tests.
5. Compare against future competitors and changing technology.
6. Ask whether the thesis identified a product instead of a deeper structural law.
7. If the thesis survives, rewrite it stronger.
8. If it partially survives, keep only the surviving kernel.
9. If it fails, abandon it and replace it.

Truth is the only objective. Never defend sunk costs.
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Prompt 4 - Civilizational Discovery

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You are not acting as an entrepreneur or strategist. You are acting as a civilizational theorist.

Assume modern civilization depends on hidden assumptions. Your task is to identify which assumptions
    are beginning to fail and what new structural laws follow.

1. Identify civilization's hidden assumptions.
2. Rank them by importance.
3. Estimate which assumptions become false, with timeframe and confidence.
4. Derive the smallest number of structural laws that explain the largest number of future changes.
5. Reconstruct major domains from first principles: law, markets, education, science, work,
    government, healthcare, media, security, cities, identity.
6. Identify missing institutions that become necessary.
7. Derive industries only after the structural laws are clear.
8. Attack the worldview.
9. Simulate 20-50 years.
10. State what would falsify the theory.

Do not optimize for drama. Optimize for coherence and predictive power.
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Prompt 5 - Reality-Constrained Audit

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The previous theory may be elegant. Your job is to check whether it is true.

Treat every attractive theory as guilty until proven otherwise.

1. Build an evidence ledger for every major claim:
   A = direct evidence
   B = strong inference
   C = weak inference
   D = unknown
   E = contradicted
2. Separate time horizons: today, 3 years, 10 years, 25 years, speculation.
3. Check current reality: deployments, businesses, regulations, incentives, bottlenecks, failures.
4. Compare with historical transitions.
5. Build competing theories.
6. Search for disconfirming evidence harder than confirming evidence.
7. Explain market reality: why does this not already exist?
8. Produce dated, falsifiable predictions.
9. Calibrate numerical confidence.
10. List residual unknowns and cheapest tests.

The goal is not the most elegant worldview. The goal is the most reality-constrained worldview.
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Prompt 6 - Discovery Engine

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Do not generate another theory. Discover the process by which accurate theories are discovered.

Reverse engineer major breakthroughs from unrelated fields: Darwin, Einstein, Shannon, Turing,
    Semmelweis, plate tectonics, germ theory, information theory, computation, economics.

For each:
1. What observations existed beforehand?
2. What assumption was abandoned?
3. What anomaly mattered?
4. What representation changed?
5. What mechanism made the theory real?
6. Why did others miss it?

Then extract a domain-general discovery process.

Test it by asking whether it could have rediscovered known breakthroughs using only pre-breakthrough
    evidence.

Final output: a research operating system, not another theory.
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Prompt 7 - Augmented Epistemology

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The goal is not to build a smarter AI. The goal is to build a better process for producing reliable
    knowledge with humans and AI together.

Design a research program for human-AI discovery.

Include:
1. Mission and north star.
2. Core principles.
3. Research loop.
4. Historical rediscovery tests.
5. Cross-domain validation.
6. Prospective predictions.
7. Failure learning.
8. Human-AI role division.
9. Metrics: prediction accuracy, calibration, compression, transferability, revision quality,
    discovery rate.
10. Confirmable success criteria.

The methodology itself is the primary hypothesis.
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3. The Core Method: Apply to Any Domain

Use this loop whenever you want to investigate a domain instead of merely asking for opinions.

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1. Observe without explaining.
2. Build an evidence ledger.
3. Define the ontology: what actually exists here?
4. Map hidden assumptions.
5. Hunt anomalies and residuals.
6. Change representation.
7. Generate competing theories.
8. Demand a mechanism.
9. Apply constraints.
10. Make dated predictions.
11. Search for disconfirming evidence.
12. Revise or kill the theory.
13. Log what failed and improve the method.
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Universal Questions

• What is the real object of study?

• What assumptions are hidden inside the words we are using?

• Where does the current explanation fail most clearly?

• What mechanism would generate the pattern?

• What alternative theory explains the same facts better?

• What would I expect to observe if this theory were false?

• What prediction can be checked within 6-36 months?

• What cheap test would reduce the most uncertainty?

4. What It Can Help With

• Business: find assumptions in an industry that are silently becoming false.

• Software: identify the hidden invariant before writing architecture.

• AI research: separate capability claims from deployment, incentives, and verification.

• Science: distinguish pattern-only findings from mechanism-backed theories.

• Forecasting: turn broad theories into dated predictions.

• Strategy: compare several world models instead of defending one narrative.

• Personal learning: train yourself to update beliefs instead of collecting elegant explanations.

5. Tests That Make It Confirmable

The method matters only if it performs better than unstructured reasoning. These tests make that measurable.

Historical Rediscovery

Give the engine only evidence available before a breakthrough. Score whether it identifies the anomaly, hidden assumption, mechanism, and eventual insight.

Prediction Log

Make 6-month, 1-year, 3-year and 5-year predictions. Review them on schedule. Track calibration.

Cross-Domain Transfer

Apply the same method to physics, biology, economics, software, psychology, law, business, and history. See what remains useful.

Compression Score

Compare theories by explanatory scope, assumptions required, predictive power, and falsifiability.

Failure Journal

For each failed prediction, record wrong assumption, wrong mechanism, wrong representation, missing constraint, and method revision.

Human vs AI vs Human-AI

Compare one human, one frontier AI, and a structured human-AI team using the same benchmark.

6. Human-AI Collaboration Protocol

A strong setup is not one model giving one answer. It is multiple roles generating disagreement, then synthesis.

- Generator: proposes candidate theories.
- Historian: checks historical parallels.
- Skeptic: tries to falsify the theory.
- Statistician: checks measurement and uncertainty.
- Economist: checks incentives and market reality.
- Engineer: checks feasibility and constraints.
- Philosopher: checks definitions and ontology.
- Prediction Auditor: demands dated, falsifiable predictions.
- Synthesizer: integrates only what survives critique.

7. Lessons and Wisdom Distilled

Reality outranks elegance.

A theory becoming smaller can be progress.

The best ideas survive attack; weak ideas survive only admiration.

Compression without prediction is storytelling.

Prediction without mechanism is coincidence.

Mechanism without evidence is speculation.

Unknowns are assets, not failures.

The highest-value assumption is often hidden inside the vocabulary.

Most discoveries begin where the current theory is embarrassed.

A smarter mind may not reason harder; it may search differently.

Attention allocation may be more fundamental than intelligence.

A bounded intelligence must choose what not to think about.

If an AI is smarter than you, judge outputs indirectly: predictions, mechanisms, reproducibility, calibration, and reality tests.

The goal is not to be right once. The goal is to become reliably less wrong over time.

The method itself is the primary hypothesis.

8. Questions Worth Pondering

• Why is reality compressible at all?

• Why do abstractions work?

• Is intelligence mainly compression, prediction, search, control, or attention allocation?

• Can discovery itself be systematically improved?

• How should finite intelligence allocate its next unit of attention?

• How can a weaker intelligence evaluate a stronger one?

• Which assumptions does civilization inherit without noticing?

• What concepts are missing because language has not needed them yet?

• What separates explanation from understanding?

• What would make this methodology fail?

9. Recommended Next Direction

Do not keep asking for bigger theories. The next move is to test the method.

• Create a prediction log with dated claims.

• Create a historical rediscovery benchmark with 10-20 cases.

• Run one case with: human alone, AI alone, and structured human-AI roles.

• Score which approach identifies anomalies, mechanisms, and false leads best.

• Revise the methodology after every failure.

The final artifact is not a theory. It is a process that improves when theories fail.

10. Minimal Repository Structure

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Augmented_Epistemology/
  00_FOUNDATIONS.md
  01_METHOD.md
  02_PROMPTS.md
  03_MENTAL_MODELS.md
  04_THEORY_LIBRARY.md
  05_FAILED_THEORIES.md
  06_OPEN_QUESTIONS.md
  07_HISTORICAL_BENCHMARKS.md
  08_PREDICTION_LOG.md
  09_EXPERIMENTS.md
  10_RESULTS.md
  11_METHOD_REVISIONS.md
```

11. Final Summary

What we learned is not that AI can magically discover truth. We learned that AI can be pushed into a disciplined epistemic workflow: generate, attack, ground, predict, falsify, revise. That workflow is more valuable than any single answer it produced.

> We began by trying to use AI to think more deeply. We ended by designing a process that helps humans and AI become more reliably wrong less often, and more reliably right over time.